

## July 2025 CSE 208

### Online on APSP

Time: 30 minutes

A logistics company manages a delivery network represented as a weighted directed graph. Each node is a city, and each edge is a direct road with a specific cost (fuel/tolls). To ensure quality control, the company has mandated that every delivery route must pass through one of their two regional distribution centers, located at **city Y and Z**, regardless of whether it is the most efficient path.

Calculate the shortest possible cost to travel between all pairs of cities  $(i,j)$  under the strict constraint that the **path must include either city Y or city Z**.

### Input

The first input line has two integers  $n$  and  $m$ : the number of cities and roads ( $1 \leq n \leq 500$ ,  $1 \leq m \leq n^2$ ).

Then, there are  $m$  lines describing the roads. Each line has three integers  $a$ ,  $b$  and  $c$ : there is a road between cities  $a$  and  $b$  whose length is  $c$  ( $1 \leq a, b \leq n$ ,  $1 \leq c \leq 10^9$ ).

Next line will take the city  $Y$  and  $Z$

Next line contains the number of queries  $q$

Finally, there are  $q$  lines describing the queries. Each line has two integers  $a$  and  $b$ : determine the length of the shortest route between cities  $a$  and  $b$ .

### Output

Print the length of the shortest route for each query. If there is no route, print -1 instead.

### Example

| Sample Input                                                                | Sample Output |
|-----------------------------------------------------------------------------|---------------|
| 4 5<br>0 2 5<br>0 1 10<br>1 2 2<br>0 3 5<br>3 2 2<br>1 3<br>2<br>0 2<br>1 2 | 7<br>2        |

|                                                                                                       |             |
|-------------------------------------------------------------------------------------------------------|-------------|
| 5 7<br>0 1 2<br>0 2 10<br>1 2 3<br>2 3 1<br>1 3 10<br>0 3 2<br>3 4 1<br>1 3<br>3<br>0 4<br>1 3<br>0 2 | 3<br>4<br>5 |
|-------------------------------------------------------------------------------------------------------|-------------|